Lab Session 5

Examine audio processing

Audio processing using matlab:

To execute audio files using MATLAB commands and to observe a human generated audio signal on time scope through Simulink. Audio signals, much like images, can undergo filtering. It is somewhat easier to understand the impact of signal processing on audio, since audio needs not be translated from a spatial to a frequency domain. To load a wave (PCM) audio file, MATLAB provides the function audioread.

funky = audioread('funky.wav');
It's important to capture the sampling frequency at which the sound was recorded; otherwise the speed of playback and result s of further processing is not guaranteed to be correct:

[funky, f] = audioread('**funky.wav**');

To play a wave file at sampling frequency f:
play(audioplayer(funky, f));
To view the waveform, plot the wave. Since audio is represented with many thousand samples per second, it may be required to plot small portions of the waveform at a time.

Spectrogram

Two-dimensional plots of audio waves can be used to easily identify magnitude; however, combined frequency distributions and magnitudes are more easily viewed in a spectrogram:
spectrogram (funky, 512 , f);
where 512 is the number of samples that are used for the discrete Fourier Transform, and thus a grouping factor of samples per column in the spectrogram image.

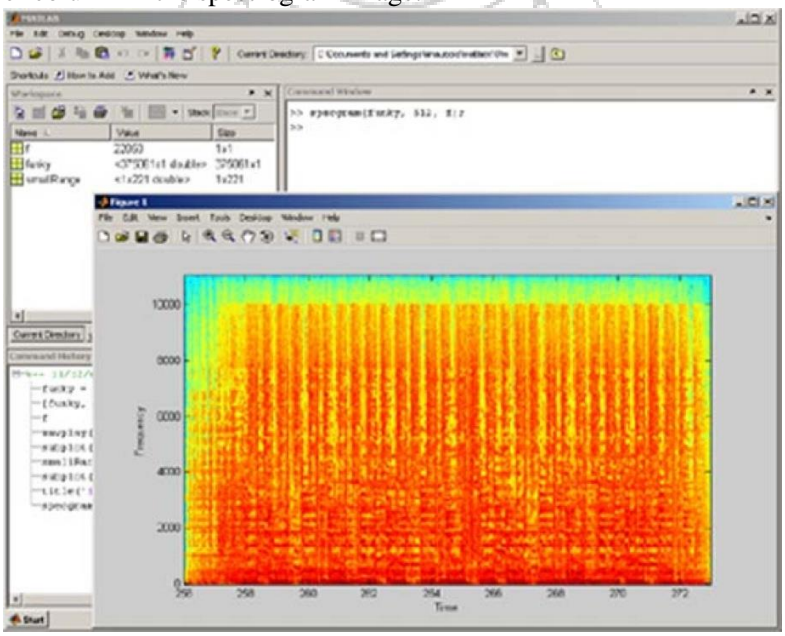


Figure 5.1: *Spectrogram*

Simulink for audio processing

Simulink is a software package for modeling, simulating, and analyzing dynamical systems. It supports linear and nonlinear systems, modeled in continuous time, sampled time, or a hybrid of the two. Systems can also be multirate, i.e., have different parts that are sampled or updated at different rates.

For modeling, Simulink provides a graphical user interface (GUI) for building models as block diagrams, using click-and-drag mouse operations. With this interface, you can draw the models just as you would with pencil and paper (or as most textbooks depict them). Simulink includes a comprehensive block library of sinks, sources, linear and nonlinear components, and connectors. To customize and create own blocks *S-Functions* can be used, which is another skill.

MATLAB and Simulink are integrated; you can simulate, analyze, and revise your models in either environment at any point.

Getting started with simulink

Opening simulink library browser:

1. In the MATLAB command window write a command "Simulink" or click on Simulink icon displayed at the toolbar. New Simulink window will be appeared click on library browser and expand these libraries to access various Toolboxes, Blocksets and Simulink functions.

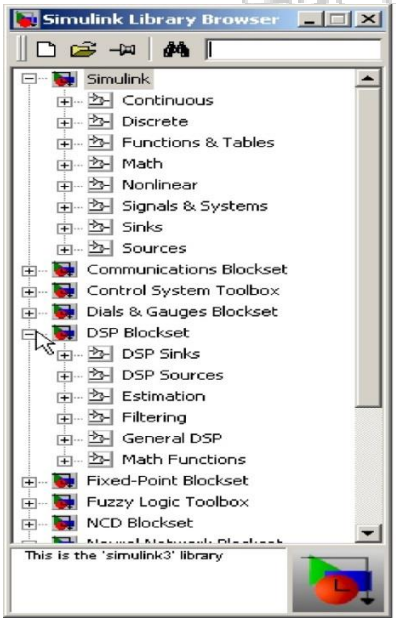


Fig. 5.2: Simulink Library Browse



Fig. 5.3: Expended Libraries

2. Write "Simulink3" in Command Window to get Simulink Block Library. This is another way of accessing Simulink Library. Double clicking each icon will open the corresponding library. Block sets and Toolboxes can be opened in the same way.

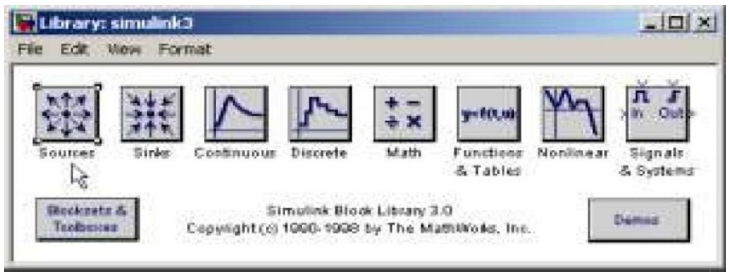


Figure. 5.4: *Simulink Block Library*

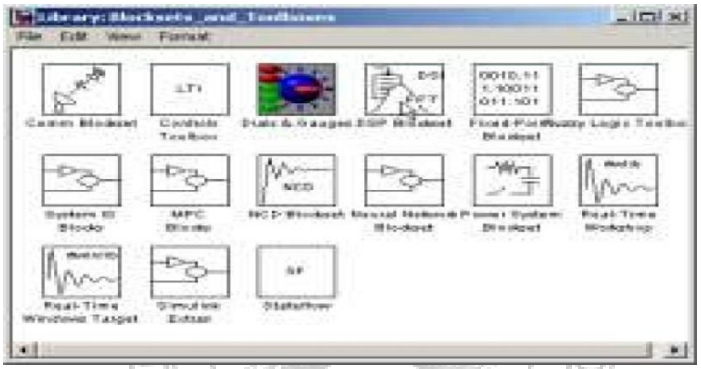


Figure. 5.5: *Simulink Library of Blocksets and Toolboxes*

Opening new simulink model:

Click to create a new model button on Simulink.

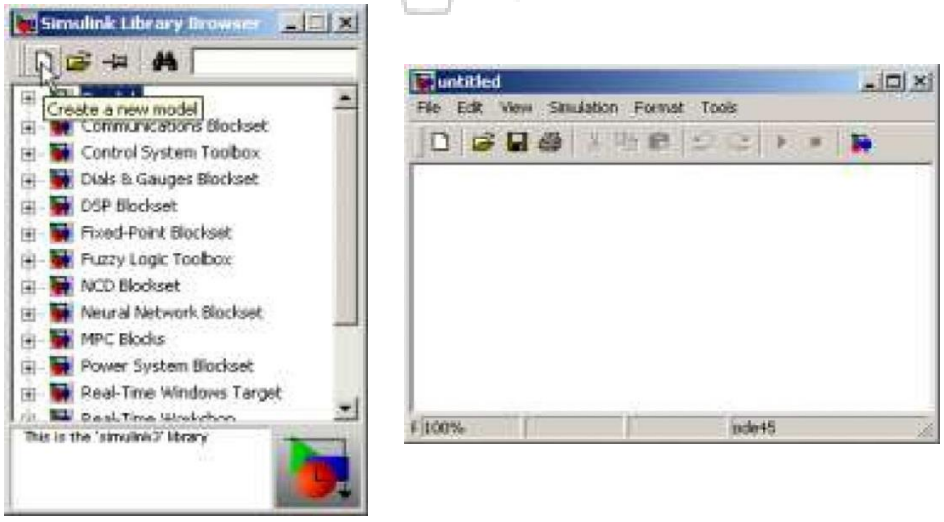


Fig. 5.6: Opening New model (.mdl file)

Human generated audio signal using simulink

The blocks used to generate a human voice in Simulink are;

Unbuffer (DSP Block set):
Convert a frame to scalar samples output at a higher sample rate.

From Wave Device (DSP Blockset):
It reads audio data samples from a standard Windows audio device in real time. Previously, only for Win95/98/NT. Now MATLAB versions support up till Windows 7 (64 bit editions included)

To Wave file (DSP Blockset):
Writes audio data samples to a standard Windows PCM format ".WAV" audio file. Previously, only for Win95/98/NT. Now MATLAB versions support up till Windows 7 (64 bit editions included).

TIME SCOPE (DSP BLOCKSET):

Provides real time monitoring of the output signal.

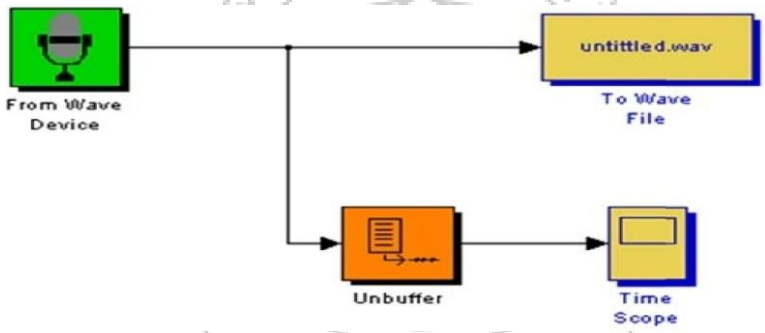


Figure 5.7: *Simulink blockset for human generated audio signal*

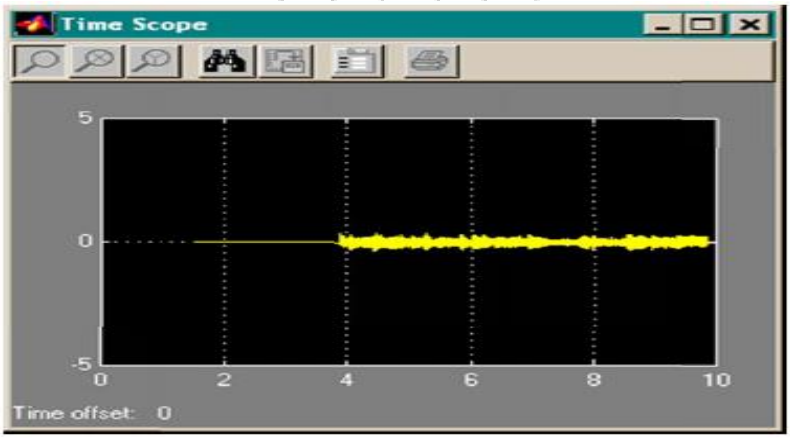


Figure 5.8 : *Time-scope view of human generated audio signal*

EXERCISE:

Task # 1: Write a MATLAB script to accomplish the following tasks:

a. Record your utterance of "we" and play it backwards. Does it sound like "you"?
b. Record your utterance of "you" and play it backwards. Does it sound like "we"?
c. Record your utterance of "We are you" and play it backwards. What does it sound like?

Task # 2: Write a MATLAB script that can read the wav file "hello.wav" and display the following information:

a. Number of sample points.
b. Sampling rate.
c. Bit resolution
d. Time duration of the recording (in terms of seconds)

Generate the output as following:

Information of Sound file "hello.wav"
Sample Points = 88200 samples
Sampling Rate = 44100 samples/sec
Bit Resolution = 16 bits/sample
Time Duration = 2 seconds

Task # 3: Write a MATLAB script to record your utterance of "today is my birthday". Try to explain the playback effect you observe after you try the following operations on the audio signals.

a. Multiply the audio signals by -1.
b. Reverse the audio signals in time axis.
c. Multiply the audio signals by 10.
d. Replace each sample by its square root.
e. Replace each sample by its square.

Task # 4: Write a MATLAB script to record your utterance with a sample rate of 32 KHz and 8-bit resolution. Try to resample the audio signals at decreasing sample rates of 16 KHz, 8 KHz, 4 KHz, 2 KHz, 1 KHz, and so on. At which sample rate you start to have difficulty in understanding the contents of the utterance?

Task # 5: Write a MATLAB script to resample the audio signals in "sunday.wav" such that new waveform has a new sample rate of 11025. Plot these two waveforms and their absolute difference by using subplot.

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